The Adoption of Cloud Computing by Small to Medium Enterprises in South Africa

A systematic literature review of 40 studies through the Technology–Organisation–Environment (TOE) lens

1 Research Problem

>90%	~60%	~40%
of SA businesses are SMEs	of the workforce	of GDP

- Cloud computing offers scalability, lower capital spending and agility.
- Yet adoption by South African SMEs remains low; most still run traditional on-premises IT.
- The barriers are technological, organisational and environmental, and have not been studied together.

2 Aim & Research Question

AIM

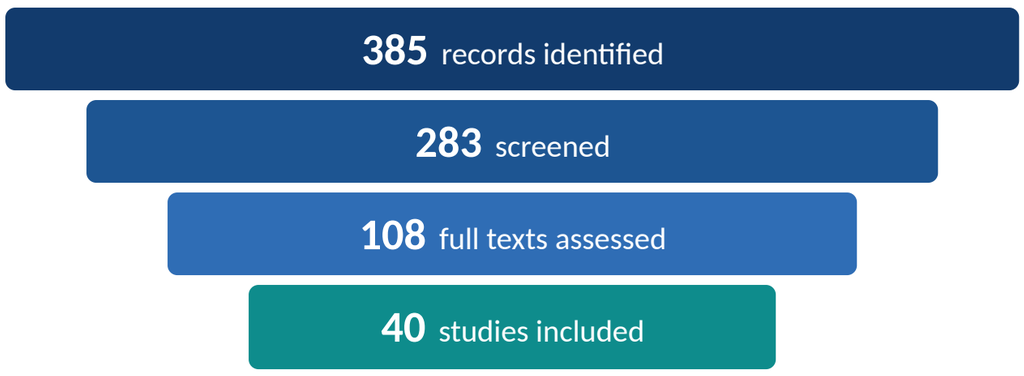
Identify the TOE factors that influence cloud adoption by South African SMEs and propose a contextualised adoption framework.

MAIN RESEARCH QUESTION

What technological, organisational and environmental factors influence cloud adoption among South African SMEs, and how can they be addressed through a contextualised framework?

3 Methodology

- Qualitative systematic literature review following **PRISMA 2020**.
- 8 databases + supplementary search + citation searching.
- Quality appraised with **MMAT**; thematic synthesis using TOE + Diffusion of Innovation factors.



36 published 2020–2026 · 35 journal articles · 37 rated high quality

4 Key Findings

STUDIES DISCUSSING EACH FACTOR (OF 40)



Technological

Security & privacy is the strongest theme. Relative advantage is the most confirmed enabler, but weakening in recent studies.

Organisational

Top management support is the most consistent driver (SA evidence mixed). The skills and preparation gap is the strongest barrier.

Environmental

Infrastructure is widely cited. Government support helps elsewhere but is seen as a barrier in SA (unclear POPIA–cloud fit).

Evidence gap: only 1 study discussed POPIA and only 2 discussed electricity or load shedding.

5 Proposed Contextualised TOE Framework

South African context layer — POPIA & data location · electricity & connectivity reliability · owner-manager awareness

Technological	Organisational	Environmental
Relative advantage · security & privacy · complexity	Top management support · skills · readiness & finance	Peer pressure · government policy · infrastructure
Response: start with SaaS; local data hosting	**Response:** owner training; prepare before migrating	**Response:** continuity plan; POPIA guidance

Enabling layer — Provider & intermediary support: vendor guidance, managed services, local partners

Informed, affordable and compliant cloud adoption by the SME

6 Conclusions & Recommendations

Cloud adoption among SA SMEs is shaped less by the technology itself than by awareness, skills, leadership commitment, regulatory clarity and infrastructure.

SME owners

Start with low-complexity SaaS, prepare properly, prefer local data residency and plan a load-shedding fallback.

Cloud providers

Transparent rand pricing, clear statements of data location and managed local support.

Policy makers

DSBD, SEDA and the Information Regulator: plain-language POPIA guidance and digital literacy for owner-managers.

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Systematic Literature Review – 40 studies, 36 published 2020–2026
PRISMA 2020 · MMAT 2018 · TOE framework